

ERROR DECOMPOSITION AND BAND-WISE IDENTIFIABILITY OF EXTREME-DARK IMAGE ENHANCEMENT

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ABSTRACT

Extreme-dark enhancement is scored by one number that does not say which part of the residual moved. We decompose a model’s residual into a global gain, a per-channel gain, a spatially varying low-frequency term and radial bands, price each with an oracle ceiling (control-corrected where spatial), and ask band by band whether the residual is information the model discards or information the rendition never carried. On a Sony sRGB benchmark the tone and colour ceiling is real but bounded, the structural term is a collapse of high-frequency correlation, and no band carries single-frame agreement the model discards beyond a small share after noise correction; on a second benchmark a different term dominates. Averaging frames raises the input’s correlation, yet the single-frame model turns the average into a larger low-frequency error. No predictor we tried recovers the single-frame block gain field. Protocol choices move the score as much as inter-method margins.

Index Terms— Low-light image enhancement, error decomposition, identifiability, evaluation protocol, image restoration

1. INTRODUCTION

Extreme-dark image enhancement is evaluated by scoring a network’s output for a short-exposure frame against a long-exposure reference with a distortion metric. Method papers report that a new architecture raises that score, often by a fraction of a decibel, and the score does not say which part of the residual the new component removed, nor whether the rest is a modelling failure or a property of the measurement; both are measurable, and the answers change where effort goes.

This paper takes reproduced models on an extreme-dark and a standard low-light benchmark, eighteen method–benchmark pairs in all, separates the residual into a global gain, a per-channel gain, a spatially varying low-frequency term and radial bands, and reports for each an oracle ceiling, control-corrected where spatial.

Ceilings say what a part is worth, not whether it is reachable, so we add a band-wise identifiability test that compares, per radial band, the model output’s correlation with the reference against the input’s own after a scalar brightness match. If the input’s own correlation is below a fixed floor, the band is not identifiable by the test; otherwise a model correlation below the input’s by more than a fixed margin marks information the model discards, and anything else is exhausted in the single-frame sense. Because the benchmark has several short frames per scene, the same test on k -frame averages separates absent information from information buried in noise.

Related work has shown that low-light enhancement can harm downstream consumers [1] and that judging it through detectors retrained on enhanced images is unreliable [2], that recent architectures add Retinex decompositions and unfoldings [3, 4], zero-reference

curves [5], normalising flows [6], colour spaces, Fourier branches, signal-to-noise gating and transformers [7, 8, 9, 10, 11, 12], and diffusion priors [13, 14], that the brightness mismatch between output and reference has been addressed by a mean-matching loss [15], that burst pipelines merge frames by noise variance in the frequency domain [16] or learn from dark bursts [17], and that diverse and challenging conditions remain the subject of a dedicated challenge [18]. Oracle ceilings have a long history in denoising [19, 20], and error diagnosis by decomposition in detection [21]. Closest to our analysis, a photometric alignment loss has been derived from a split of the residual into a global photometric part and a structural part [22]; what we add is an oracle ceiling per component, control-corrected for the spatially varying ones, and the band-wise identifiability test. We propose that decomposition and that test as a method for locating an enhancement model’s remaining error, and report what they find.

2. PROPOSED METHOD

We propose a measurement procedure with two parts: a decomposition of a model’s residual into named components, each priced by an oracle ceiling, and a band-wise identifiability test that says whether the residual in a band is information the model discards or information the rendition never carried. Let y be the model output and g the reference, both in $[0, 255]$ per channel. Scores and correlations are computed per test frame and averaged; error shares are pooled over frames; block ceilings are PSNR without clipping so that a control can be subtracted; the gain terms are scored like the model.

Gain terms. The global term is the least-squares scalar $a^* = \langle y, g \rangle / \langle y, y \rangle$ applied to the whole frame and the per-channel term repeats it per colour channel; their ceilings are the scores of a^*y and of the per-channel fit, and the share of squared error each removes is the tone and colour part of the residual.

Spatially varying low-frequency term. We partition the frame into square blocks of side b and fit one scalar per block, $y \mapsto a_{\text{blk}}y$, and separately an affine map $y \mapsto a_{\text{blk}}y + b_{\text{blk}}$. Decreasing b adds parameters and improves the fit even when there is nothing to fit, so for each b we repeat the procedure against a structureless-noise target of the same variance and report the gain over the $b = \text{frame anchor}$ of the same family after subtracting that control.

Band terms. We take the two-dimensional spectrum of the luminance and group frequencies into radial bands. For band j we report the output-to-reference power ratio and the correlation ρ_j restricted to that band; under an optimal per-band amplitude rescaling the surviving fraction of band energy is $1 - \rho_j^2$, so ρ_j , not the power ratio, bounds any amplitude correction.

Identifiability. Relative to the estimator class of a scalar brightness match followed by per-band rescaling, we compute the same ρ_j between the reference and the input itself, after matching brightness with a single scalar so that no structure is altered, and call it ρ_j^{in} . With the two thresholds fixed in the measurement code released with the

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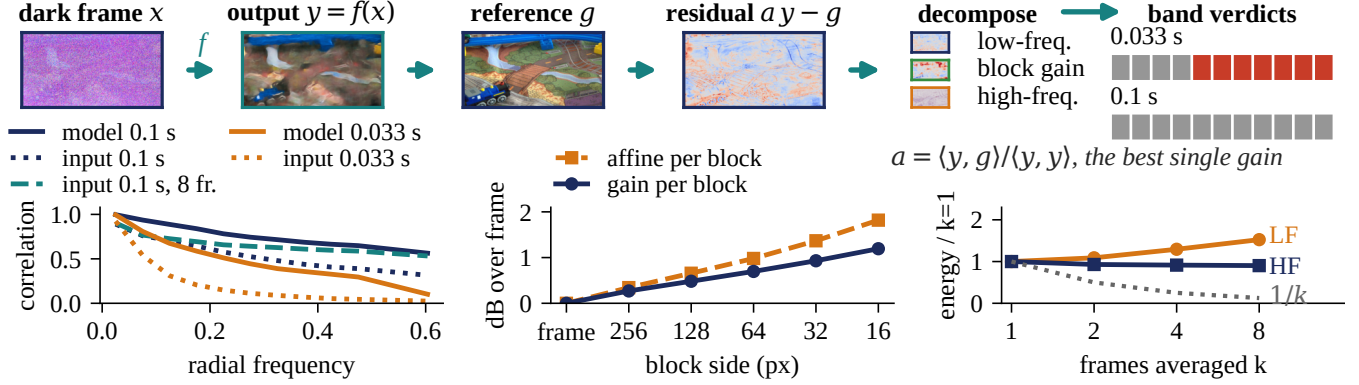


Fig. 1. Overview, worst 0.033 s Sony frame (input $\times 20$): the residual after the oracle global gain is split into low-frequency, per-block gain and high-frequency parts (real maps from this frame), and every radial band is labelled by whether the input still carries its structure (red: unidentifiable, grey: exhausted; green would mark recoverable; no band reaches it on this frame). Bottom: the three measurements over the benchmark (LF/HF: low-/high-frequency error energy; the 0.1 s input curves use the 36 groups with at least eight frames).

paper, a band is *unidentifiable* when $\rho_j^{\text{in}} < 0.20$, so the test cannot certify structure recovered there even where the model’s correlation exceeds the floor (Table 2); *recoverable* when $\rho_j^{\text{in}} - \rho_j > 0.05$, so the band carries agreement the model discards; and *exhausted* otherwise. This raw-input check is necessary but easy to pass, so we also report a noise-corrected ceiling: the input’s noise power N_j per band is measured from frame-to-frame differences and ρ_j^{in} is corrected to $\rho_j^{\text{in}} / \sqrt{1 - N_j/P_j}$, P_j the input band power, which assumes additive noise uncorrelated with the reference; cells with $N_j/P_j > 0.9$ are unstable under this correction and are reported as such.

Predictability of the local term. A ceiling obtained with the reference in hand is not a method, so for the block term we ask whether the oracle coefficients can be predicted at inference: we regress the within-frame centred block coefficients on the input block mean, the input block variance and the output block mean, cross-fitting by scene so that no frame of a scene is used to fit its own prediction, and as a reference point repeat the fit with the reference block mean included, which bounds what a block-level summary can explain.

3. EXPERIMENTAL RESULTS

Setup. Every table and figure applies the procedure of Sec. 2 to released checkpoints or training-free baselines; the reproduction is in Sec. 4. We use the Sony sRGB extreme-dark benchmark derived from SID [23] in the processed sRGB form prepared for SNR-Net [11] and released with Retinexformer [24]: 50 test scenes and 598 short-exposure frames at 512×960 , each scored against the single long-exposure frame of its scene. As a standard low-light benchmark we use LOL [25] (15 test frames) with its v2 sets [26]. The model under analysis is the released Retinexformer checkpoint; inference is deterministic, so every spread is over frames or scenes. We reproduce its reported score to within 0.002 dB with our own metric implementation (Sec. 4).

Existing methods under one protocol, and the proposed decomposition applied to each. Table 1 scores, with rounding and no rectification, the outputs of 11 methods on LOL and 7 on Sony: CLAHE [27], the zero-reference Zero-DCE++ [5], SCI [28] (released weights trained on LOL and LSRW), an in-house self-supervised Retinex model (AR), the unsupervised diffusion model LightenDiffusion [14] (unpaired training on LSRW, not evaluated on Sony by its authors), and networks trained on each benchmark with its reference (URetinex-Net [4], SNR-Net [11], LLFormer [12], the diffusion model GSAD [13], Retinexformer, CIDNet [7]); the rightmost four columns are Sec. 2 applied to each method’s own

residual; Fig. 2 shows one frame. On LOL six methods clear 20 dB unrectified: the reference-trained networks other than URetinex-Net with its fixed ratio, and LightenDiffusion at 20.12, a count that its sampling spread can lower to five; the rest stay below 20 dB. On Sony only the two trained on it, SNR-Net and Retinexformer, clear 20 dB, and LightenDiffusion leaves 24 percent of its error in the per-channel term, a colour cast. Across the enhancement methods in Table 1 (bounds rounded outward) the global term ranges from 41 to 89 percent of the error on LOL and from 14 to 42 on Sony, so which term dominates varies with method and benchmark, while the block ceiling stays between 0.8 and 3.0 dB on LOL and 0.7 and 1.8 dB on Sony: the spatially varying low-frequency term is common to all.

Global tone and colour: a priced ceiling, not an exhausted one. On Sony an oracle global gain and per-channel gain account for 27.0 and 8.8 percent of Retinexformer’s squared error, taking the score from 24.438 to 26.484 dB; the remaining 64.2 percent survives any global or per-channel gain, growing from 60.8 percent at 0.1 s to 73.0 at 0.033 s.

The structural term is decorrelation, not attenuation. Output power falls with radial frequency (ratio 0.994 in the lowest band, 0.279 at the corner), and the output is closer to a blurred reference than to the reference: with $\sigma = 0.6$ the output scores 26.661 dB against the blurred reference, 0.882 dB above its gain-corrected score against the reference (25.779), which is the signature of a conditional-mean solution [29]. Restoring the lost amplitude buys almost nothing: rescaling each band of the output to the reference power with its own phase moves the score by 0.440 dB, the least-squares per-band gain by 0.947 dB, because ρ_j falls from near unity in the lowest band to 0.102 at the corner for the shortest exposure.

Two readings, stated together. Against the brightness-matched input the model’s correlation is at or above it in every band and at every exposure (Table 2), so no band is recoverable under that criterion; at 0.033 s 24.4 percent of the error sits in bands where the input’s own agreement is below the floor. Under the noise-corrected ceiling of Sec. 2 the recoverable share is 0.0, 4.5 and 3.0 percent at 0.033, 0.04 and 0.1 s. Corrected cells with $N_j/P_j > 0.9$, the bands above normalised frequency 0.20 at 0.033 s and above 0.35 at 0.04 s, are read as unstable; the 0.04 s share of 4.5 percent comes entirely from one such cell ($N_j/P_j = 0.956$), the 0.1 s share from a stable one. The verdict is margin-sensitive: at margin zero the three shares become 6.5, 21.9 and 3.0 percent, 6.5 and 14.9 points of which come from those unstable cells, and the two bands that clear the 0.05 margin exceed it by 0.013 and 0.005. The additive-noise assumption

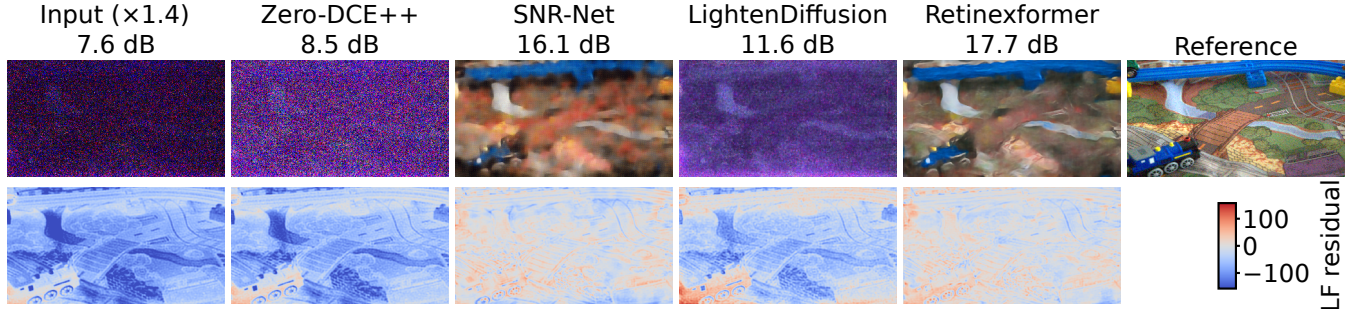


Fig. 2. Existing methods on the 0.033 s Sony frame on which Retinexformer scores lowest: input after its oracle global gain (label: score before it), four outputs, and the reference (top); the decomposition applied to input and methods: the low-frequency part (below 0.10) of each residual after its own oracle global gain (bottom, common 8-bit range). Every method leaves a spatially varying brightness field of method-specific size.

behind the correction was tested on the same captures: on this scalar path it passes the aggregate checks and fails the lowest-band check at 0.04 and 0.1 s (not run at 0.033 s, two frames per scene), where the noise estimate is small and the output correlation exceeds 0.99, so no verdict can change. It fails at low frequency on the alternative pointwise path, which fits a per-frame tone curve to the reference and hands the input information it does not have; there the shares rise to 6.5, 21.9 and 19.3 percent, an optimistic bound.

This single-frame checkpoint cannot consume a burst. The natural way to spend the multi-frame margin is to average k input frames. On the 50 scene–exposure groups with at least eight frames (36 at 0.1 s, 14 at 0.04 s), so that every k is scored on identical groups, this raises the low-frequency error energy after the oracle global gain monotonically, to 1.089, 1.294 and 1.524 times the single-frame value at $k = 2, 4$ and 8 , where pure variance would have given $1/k$, while the high-frequency energy falls only to 0.903. On the 36 scenes at 0.1 s the raw score of the eight-frame mean falls by 4.57 dB, yet on the same 36 groups the input’s own band correlation (all 361 frames of those groups, then their 36 eight-frame means) rises in every band, at the corner from 0.315 to 0.532, and the model’s on the averaged input (eight frames per group) in nine of eleven (the lowest two fall), at the corner from 0.583 to 0.724, so the structure improves and the brightness field collapses. Averaging the model’s outputs instead of its inputs gains 0.672 dB at $k = 8$; the noise-realisation share (half the mean squared difference between outputs of two frames of one scene and exposure over the model’s squared error) is 10.6 percent, 0.49 dB, on the full test set under an independence assumption. A noise-variance frequency-domain burst merge [16], on all 50 groups at $k = 8$ with the noise level estimated from the frames, scores 21.71 dB against 20.81 (plain mean) and 24.69 (one frame): it recovers 0.91 dB of what the mean loses, almost all below 0.05 normalised frequency, and stays 2.97 dB below a single frame. The dominant term is a frame-common bias, not the model reading noise level as a brightness cue: re-injecting noise into the eight-frame mean moves the low-frequency error to 1.775 and adding noise to a single frame to 1.406; the checkpoint’s low-frequency estimate is tuned to its training statistics and departs from the reference either way.

What carries over to the second benchmark. Three findings carry over from Sony to LOL: the output again closer to a blurred reference; the low-frequency part again dominant, 67 to 70 percent of the error energy; and the block oracle again above a decibel, 2.29 and 1.17 dB at 16 pixels (Δ_{16} in Table 1), of which a control fitted to another frame’s residual takes 1.664 and 0.858 dB, 73 and 73 percent of them against 99 percent on Sony, so about a quarter of the LOL block ceiling is frame-specific, on Sony almost none. Hygiene check: all eleven LOL bands are exhausted against the raw input for both

CIDNet variants and Retinexformer.

The dominant term is a spatially varying low-frequency field. The residual after the global gain is concentrated at low frequency: the low-pass part carries 64.4 percent of the error energy, of which a single per-frame constant explains only 8.8 percent. Fig. 1 prices it: an oracle allowed one scalar per 16-pixel block gains 1.194 dB over the whole-frame fit, an offset per block 1.09 dB, and an affine map per block 1.819 dB, after subtracting the structureless-target control, which is negligible. A second, more demanding control, fitting the same block gains to another frame’s residual of matched variance, recovers 1.185 dB, 99 percent of the frame’s own control-corrected value. The block oracle therefore prices a spatially varying low-frequency correction in general, not this frame’s own structure; the 64.4 percent share is energy, the block ceiling is what one scalar per block removes of it.

This term is large but, by the predictors we tried, not visible from the input. The three-statistic regression of Sec. 2, cross-fitted by scene, explains a fraction 0.006 of the coefficients’ variance; including the reference block mean, which is not available at inference, raises it to 0.693. Three scalars are a weak test, so we also trained a convolutional predictor on the whole input frame, split by scene, with the rule fixed in advance that R^2 below 0.1 and a gain below 0.15 dB count as not predictable; it reaches $R^2 = 0.019$ on held-out scenes and, applied to the output, 0.010 dB, 0.8 percent of the oracle ceiling of the same block size. It is a small U-Net trained on the benchmark’s 161 training scenes, with a label-shuffled control at or below zero and a single-batch overfit to $R^2 = 0.87$; the predictors we tried do not recover the field from this rendition, which does not show that none could.

4. DISCUSSION

Reproduction. Every baseline we quote was rerun by us and scored with our own metric implementation; independent PSNR and SSIM implementations agree with the released ones to machine precision. For CIDNet and Retinexformer the reproduced scores agree with the published ones to 0.021 dB except one cell at 0.081, a gap we could not attribute (published/here: CIDNet LOL 23.50/23.498, with perceptual loss 23.8091/23.808, LOL-v2 real 28.1387/28.140 and synthetic 29.5663/29.647 rectified; Retinexformer LOL 27.18/27.169 rectified, Sony 24.44/24.438, LOL-v2 real 27.71/27.689 and synthetic 29.04/29.022 rectified); SNR-Net, LLFormer and URetinex-Net in its reference-ratio setting reproduce their published LOL scores to 0.003 dB, the two diffusion models to 0.13 dB (GSAD, under rectification, beyond its single-run spread) and 0.34 dB (LightenDiffusion, within its seed spread), and Zero-DCE++ has no published LOL score; this is what licenses reading the decomposition as a property of the method.

Table 1. Existing methods under one protocol (rounding, no rectification) and the decomposition of each residual (rightmost four columns): PSNR in dB (mean $\pm$ sd over frames; 15 on LOL, 598 on Sony), SSIM, squared-error shares in percent (pooled over frames) removed by the global gain, the per-channel gain and the remainder, and Δ_{16} in dB (frame mean), the control-corrected 16-pixel block-gain ceiling over the whole-frame gain. Diffusion models: one sample, fixed seed (LightenDiffusion’s seed spread 0.57 dB). URetinex-Net: fixed ratio (its published score uses a reference-derived ratio). GSAD: published 27.84 is rectified (27.706 here). Zero-DCE++: run on the 12-multiple crop its code requires, curve map edge-extended. AR: in-house Retinex model, self-supervised on COCO. Gamma 0.5 and the perceptual-loss CIDNet variant appear only in the released comparison table. Rows: by benchmark; reference-free methods first, then reference-trained ones by year; Sony rows: methods with a Sony checkpoint or none needed. LOL per-frame sd: shares up to 34 points, Δ_{16} up to 2.2 dB.

method	PSNR	SSIM	gl.	ch.	res.	Δ_{16}
<i>LOL</i>						
Input	7.77 $\pm$ 2.62	0.195	88.2	0.7	11.2	3.15
CLAHE	9.13 $\pm$ 2.96	0.362	88.2	0.4	11.4	2.99
Zero-DCE++	15.34 $\pm$ 4.66	0.567	75.7	3.4	20.9	0.88
SCI	14.85 $\pm$ 4.43	0.524	70.1	2.1	27.8	1.74
AR (in-house)	11.04 $\pm$ 3.72	0.449	86.5	0.8	12.7	0.96
LightenDiffusion	20.12 $\pm$ 3.09	0.813	62.7	3.6	33.7	1.87
URetinex-Net	19.86 $\pm$ 3.97	0.825	73.5	3.2	23.2	1.73
SNR-Net	24.61 $\pm$ 4.26	0.842	53.7	4.3	42.0	1.15
LLFormer	23.65 $\pm$ 4.63	0.818	63.9	3.0	33.2	1.13
GSAD	22.81 $\pm$ 5.42	0.852	80.9	1.3	17.8	1.37
Retinexformer	25.15 $\pm$ 2.87	0.845	41.4	4.1	54.5	1.17
CIDNet	23.61 $\pm$ 5.44	0.870	75.7	1.1	23.3	2.29
<i>Sony</i>						
Input	11.50 $\pm$ 2.82	0.150	21.7	4.6	73.7	1.14
CLAHE	13.32 $\pm$ 3.08	0.223	26.2	4.1	69.7	1.22
Zero-DCE++	12.74 $\pm$ 3.32	0.138	28.4	6.3	65.3	1.27
SCI	11.80 $\pm$ 3.22	0.138	41.4	4.4	54.3	1.02
AR (in-house)	12.93 $\pm$ 3.33	0.148	15.0	3.5	81.5	0.75
LightenDiffusion	17.09 $\pm$ 3.54	0.461	27.3	23.9	48.8	1.71
SNR-Net	22.87 $\pm$ 3.52	0.625	33.6	9.9	56.5	1.35
Retinexformer	24.44 $\pm$ 3.51	0.680	27.0	8.8	64.2	1.19

Three protocol choices move the score by as much as the margins do. First, reference-mean rectification sets the output’s brightness from the reference. On the Sony benchmark it is worth 1.16 dB, the whole-frame special case of the spatially varying term priced in Sec. 3, and for GSAD on LOL 4.90 dB; a related setting, URetinex-Net’s reference-derived ratio, is worth 1.47 dB (21.33 published against 19.86 at its fixed ratio, Table 1). Retinexformer’s documentation recommends against the setting while CIDNet reports its headline number with it, so the two are not comparable; the latter’s script further scales LOL-v2 outputs by a per-checkpoint constant chosen against the reference (about three decibels unrectified). All three are reasons to rerun baselines, not criticisms of a method. Second, the 8-bit conversion rule: one pipeline saves outputs by truncation and the other by rounding, confirmed at the pixel level on saved files. Switching the rule alone is worth between 0.031 and 0.110 dB on the three LOL checkpoints. Third, the dataset variant: the Sony extreme-dark set is distributed in at least two renditions whose pixel conventions differ [24, 7, 8], and the checkpoint of [7] scores 5.29 dB on the other (598 frames, rounding) against the 22.904 reported for it on its own.

What the decomposition implies for method design. The headroom on Sony is 1.34 dB for a global gain and 0.71 dB more for a per-

Table 2. Band-wise correlation with the reference: model output against scalar brightness-matched input (Sec. 2 primary path, all frames); conservative in the model’s favour, the text gives the noise-corrected reading; † marks unstable input cells ($N_j/P_j > 0.9$).

band	0.033 s ($n = 28$)		0.04 s ($n = 180$)		0.1 s ($n = 390$)	
	model	input	model	input	model	input
0.00–0.05	0.995	0.923	0.996	0.888	0.997	0.891
0.05–0.10	0.809	0.541	0.858	0.609	0.937	0.756
0.10–0.15	0.674	0.310	0.753	0.454	0.888	0.699
0.15–0.20	0.587	0.207	0.672	0.346	0.839	0.632
0.20–0.25	0.511	0.148†	0.595	0.261	0.782	0.560
0.25–0.30	0.444	0.112†	0.529	0.206	0.743	0.503
0.30–0.35	0.390	0.089†	0.483	0.169	0.716	0.456
0.35–0.40	0.357	0.069†	0.449	0.138†	0.686	0.411
0.40–0.45	0.327	0.053†	0.417	0.114†	0.664	0.379
0.45–0.50	0.295	0.045†	0.392	0.102†	0.648	0.360
0.50–0.71	0.102	0.025†	0.275	0.071†	0.564	0.284

channel gain, 1.19 dB for a per-block gain and 1.82 dB for a per-block affine map at 16 pixels over the whole-frame fit (both control-corrected), 0.95 dB for per-band amplitude, and about 0.49 dB for the noise share estimated under independence. The rule that follows is to spend effort on the component with the largest headroom, testing its predictors before building on it: on Sony the spatially varying low-frequency term, on LOL, by share of squared error, the global term for every method but Retinexformer (Table 1). Single-frame high-frequency restoration is worth a few percent under the noise-corrected ceiling, and forcing amplitude up makes it worse. The low-frequency term resists the predictors we tried; the multi-frame margin exists in the input but this single-frame model cannot take it, so taking it would require training on burst statistics. A testable prescription: train on k -frame means and test whether the $k = 8$ low-frequency energy drops below the single-frame value.

Limitations. All quantities are measured on the 8-bit sRGB rendition the benchmarks distribute, so every recoverability statement is about that rendition, not sensor data. Quantisation to 8 bits, and sensor noise that is not additive [30], depart from the additive uncorrelated-noise model behind the noise correction; the check of Sec. 3 fails in the lowest band at 0.04 and 0.1 s, and cells with $N_j/P_j > 0.9$ are reported as unstable. The identifiability test is a distortion-axis statement about conditional-mean regressors [29], so a perceptually optimised method can fall below the input on ρ_j without being wrong. The reference’s noise floor was not measured and is treated as zero. The band-wise test covers Retinexformer on Sony and CIDNet and Retinexformer on LOL, while Table 1 decomposes every method; CIDNet’s Sony residual was not decomposed because the variant it was trained on, in a different pixel convention, is not the one we could download. The noise-corrected ceiling exists only on Sony, which has repeated captures; band correlations are global per frame, so a band can be labelled exhausted while local disagreement remains; the multi-frame result uses static scenes only.

5. CONCLUSION

We decomposed the residual of reproduced enhancement models into components with oracle ceilings, control-corrected where spatial, and tested band by band whether the error is discarded information or information the rendition never carried. On both benchmarks the residual after the global gain sits at low frequency, a spatially varying correction of it is worth more than a decibel, and the output is closer to a blurred reference than to the reference. That term resists the predictors we tried, this single-frame checkpoint cannot consume a burst, and code and numbers are released.

6. REFERENCES

- [1] Xu Wu, Zhihui Lai, Jie Zhou, Can Gao, Xianxu Hou, Ya-nan Zhang, and Linlin Shen, “Low-light enhancement effect on classification and detection: An empirical study,” *arXiv preprint arXiv:2409.14461*, 2024.
- [2] Mingjia Li, Hao Zhao, and Xiaojie Guo, “LIME-Eval: Rethinking low-light image enhancement evaluation via object detection,” *arXiv preprint arXiv:2410.08810*, 2024.
- [3] Yonghua Zhang, Jiawan Zhang, and Xiaojie Guo, “Kindling the darkness: A practical low-light image enhancer,” in *ACM Multimedia*, 2019, pp. 1632–1640.
- [4] Wenhui Wu, Jian Weng, Pingping Zhang, Xu Wang, Wenhan Yang, and Jianmin Jiang, “URetinetx-Net: Retinex-based deep unfolding network for low-light image enhancement,” in *CVPR*, 2022, pp. 5891–5900.
- [5] Chongyi Li, Chunle Guo, and Chen Change Loy, “Learning to enhance low-light image via zero-reference deep curve estimation,” *IEEE TPAMI*, vol. 44, no. 8, pp. 4225–4238, 2022.
- [6] Yufei Wang, Renjie Wan, Wenhan Yang, Haoliang Li, Lap-Pui Chau, and Alex C. Kot, “Low-light image enhancement with normalizing flow,” in *AAAI*, 2022, vol. 36, pp. 2604–2612.
- [7] Qingsen Yan, Yixu Feng, Cheng Zhang, Guansong Pang, Kangbiao Shi, Peng Wu, Wei Dong, Jinqiu Sun, and Yanning Zhang, “HVI: A new color space for low-light image enhancement,” in *CVPR*, 2025, pp. 5678–5687.
- [8] Qingsen Yan, Kangbiao Shi, Yixu Feng, Tao Hu, Peng Wu, Guansong Pang, and Yanning Zhang, “HVI-CIDNet+: Beyond extreme darkness for low-light image enhancement,” *arXiv preprint arXiv:2507.06814*, 2025.
- [9] Jie Huang, Yajing Liu, Feng Zhao, Keyu Yan, Jinghao Zhang, Yukun Huang, Man Zhou, and Zhiwei Xiong, “Deep Fourier-based exposure correction network with spatial-frequency interaction,” in *ECCV*, 2022, pp. 163–180.
- [10] Chenxi Wang, Hongjun Wu, and Zhi Jin, “FourLLIE: Boosting low-light image enhancement by Fourier frequency information,” in *ACM Multimedia*, 2023, pp. 7459–7469.
- [11] Xiaogang Xu, Ruixing Wang, Chi-Wing Fu, and Jiaya Jia, “SNR-aware low-light image enhancement,” in *CVPR*, 2022, pp. 17714–17724.
- [12] Tao Wang, Kaihao Zhang, Tianrun Shen, Wenhan Luo, Bjorn Stenger, and Tong Lu, “Ultra-high-definition low-light image enhancement: A benchmark and transformer-based method,” in *AAAI*, 2023, vol. 37, pp. 2654–2662.
- [13] Jinhui Hou, Zhiyu Zhu, Junhui Hou, Hui Liu, Huanqiang Zeng, and Hui Yuan, “Global structure-aware diffusion process for low-light image enhancement,” in *NeurIPS*, 2023.
- [14] Hai Jiang, Ao Luo, Xiaohong Liu, Songchen Han, and Shuaicheng Liu, “LightenDiffusion: Unsupervised low-light image enhancement with latent-Retinex diffusion models,” in *ECCV*, 2024.
- [15] Jingxi Liao, Shijie Hao, Richang Hong, and Meng Wang, “GT-Mean loss: A simple yet effective solution for brightness mismatch in low-light image enhancement,” in *ICCV*, 2025.
- [16] Samuel W. Hasinoff, Dillon Sharlet, Ryan Geiss, Andrew Adams, Jonathan T. Barron, Florian Kainz, Jiawen Chen, and Marc Levoy, “Burst photography for high dynamic range and low-light imaging on mobile cameras,” *ACM TOG*, vol. 35, no. 6, pp. 192:1–192:12, 2016.
- [17] Ahmet Serdar Karadeniz, Erkut Erdem, and Aykut Erdem, “Burst photography for learning to enhance extremely dark images,” *IEEE TIP*, vol. 30, pp. 9372–9385, 2021.
- [18] George Ciubotariu, S M A Sharif, Abdur Rehman, Fayaz Ali Dharejo, Rizwan Ali Naqvi, Marcos V. Conde, Radu Timofte, et al., “Low light image enhancement challenge at NTIRE 2026,” in *CVPR Workshops*, 2026.
- [19] Priyam Chatterjee and Peyman Milanfar, “Is denoising dead?,” *IEEE TIP*, vol. 19, no. 4, pp. 895–911, 2010.
- [20] Anat Levin and Boaz Nadler, “Natural image denoising: Optimality and inherent bounds,” in *CVPR*, 2011, pp. 2833–2840.
- [21] Derek Hoiem, Yodsawalai Chodpathumwan, and Qieyun Dai, “Diagnosing error in object detectors,” in *ECCV*, 2012, pp. 340–353.
- [22] Mingjia Li, Tianle Du, Hainuo Wang, Qiming Hu, and Xiaojie Guo, “On the global photometric alignment for low-level vision,” *arXiv preprint arXiv:2604.08172*, 2026.
- [23] Chen Chen, Qifeng Chen, Jia Xu, and Vladlen Koltun, “Learning to see in the dark,” in *CVPR*, 2018, pp. 3291–3300.
- [24] Yuanhao Cai, Hao Bian, Jing Lin, Haoqian Wang, Radu Timofte, and Yulun Zhang, “Retinexformer: One-stage Retinex-based transformer for low-light image enhancement,” in *ICCV*, 2023.
- [25] Chen Wei, Wenjing Wang, Wenhan Yang, and Jiaying Liu, “Deep Retinex decomposition for low-light enhancement,” in *BMVC*, 2018.
- [26] Wenhan Yang, Wenjing Wang, Haofeng Huang, Shiqi Wang, and Jiaying Liu, “Sparse gradient regularized deep Retinex network for robust low-light image enhancement,” *IEEE TIP*, vol. 30, pp. 2072–2086, 2021.
- [27] Karel Zuiderveld, “Contrast limited adaptive histogram equalization,” in *Graphics Gems IV*, pp. 474–485. Academic Press, 1994.
- [28] Long Ma, Tengyu Ma, Risheng Liu, Xin Fan, and Zhongxuan Luo, “Toward fast, flexible, and robust low-light image enhancement,” in *CVPR*, 2022.
- [29] Yochai Blau and Tomer Michaeli, “The perception-distortion tradeoff,” in *CVPR*, 2018, pp. 6228–6237.
- [30] Kaixuan Wei, Ying Fu, Jiaolong Yang, and Hua Huang, “A physics-based noise formation model for extreme low-light raw denoising,” in *CVPR*, 2020, pp. 2758–2767.